

Jeremy Lu

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Experience

Data Analyst, NYU Langone Health Rapid RCT Lab – New York, NY May 2023 – Present

- Present quality improvement trial ideas and results to leadership teams ranging from C-Suite, kidney transplant, antibiotic stewardship, and other areas of hospital.
- Assessed relationship between high heat days and 200,000 emergency department visits over past four years using case crossover design, in order to build new patient safety messaging system.
- Evaluated 4200 encounters over 15 months from an antibiotic stewardship program using difference in differences methods and found that it reduces 30 day readmission rate by 20%.
- Built gravity model using six years of historical data totaling over a million emergency department visits to evaluate catchment area and inform patient capacity and recruitment decision making in NYC neighborhoods.

AI Research Intern, Aizip, Inc. – Cupertino, CA June 2022 – Aug 2022

- Automated process for deploying imagenet-based PyTorch Convolutional and Recurrent Neural Networks on hardware.
- Managed 3 undergraduate interns to collect, label, and organize visual and audio data for multiple TinyML neural networks in IoT.

Research Assistant, UW Department of Atmospheric Sciences – Seattle, WA Sept 2019 – Jun 2021

- Performed exploratory analysis on over 140,000 particles from Southern Ocean Clouds, Radiation, Aerosol Transport Experimental Study (SOCRATES) probe data using matplotlib, numpy, and xarray.
- Used Scikit-Learn to build and optimize a binary random forest classifier, distinguishing over 5 million liquid and ice cloud particles with 95% accuracy, a 20% increase from previous non-ML methods.

Technologies and Skills

Languages: Python (pandas, matplotlib, plotly, numpy, scikit-learn, seaborn, pytorch, onnx, flask), R (base, tidyverse, ggplot, Shiny, glm, stan, bartcause), Java, HTML, CSS, SQL, SQLite

Software: Jupyter Notebook, Amazon Web Services, Git, Microsoft Office, LaTeX, Adobe Acrobat, Tableau, Databricks, Visual Studio Code, RStudio, Docker, OpenAI, Llama, Linux

Skills: Causal Inference, Predictive Modeling, Machine Learning, Randomized Controlled Trials, A/B Testing, Experimental Design, Dashboarding and Reporting, Prompt Engineering, Behavioral Science (COM-B Framework)

Education

New York University, M.S. in Applied Statistics for Social Science Research Sept 2021 – May 2023

University of Washington - Seattle, B.S. in Applied and Computational Mathematical Sciences (Data Science Concentration) Sept 2018 – Jun 2021

Publications and Conferences

The University of Washington Ice-Liquid Discriminator (UWILD) improves single-particle phase classifications of hydrometeors within Southern Ocean clouds using machine learning [10.5194/amt-14-7079-2021](https://doi.org/10.5194/amt-14-7079-2021) Nov 2021

Rachel Atlas, Johannes Mohrmann, Joseph Finlon, **Jeremy Lu**, Ian Hsiao, Robert Wood, Minghui Dao

Optimizing Provider Decision-Making with Clinical Decision Support Tools and Behavioral Nudges, 2024 UPenn Nudges in Health Care Symposium Sep 2024

Jeremy Lu, Sarah Tsuruo, Holly Krelle, Nate Klapheke, William King, Kyra Rosen, Leora Horwitz

Behaviorally-Informed Text Message Reminders to Increase Completion of Colorectal Cancer Screenings Among Patients at the NYU Family Health Centers, 2025 UPenn Nudges in Health Care Symposium Sep 2025

William King, Cathy Jian, Olivia Larsson, **Jeremy Lu**, Sarah Tsuruo, , Holly Krelle, Nathan Klapheke, Arielle Elmalech-Sachs, Elan Pelegri, Leora Horwitz